

AMC 12 COMPREHENSIVE REFERENCE

AMC 12 Cheat Sheet

Full high-school competition toolkit — algebra, trigonometry, geometry, complex numbers, number theory, combinatorics and non-calculus optimization.

CORE instant recall

HIGH high-yield

BONUS late-paper tool

Exam map CORE

- 25 multiple-choice problems • 75 minutes.
- Scoring: **+6 correct, +1.5 blank, 0 incorrect**.
- No calculator. Figures are not necessarily drawn to scale.
- Full high-school mathematics **except calculus**.
- AMC 12 adds trigonometry, logarithms, complex numbers and stronger algebra/geometry beyond AMC 10.

Universal contest moves HIGH

- Backsolve from choices.
- Normalize scale-invariant problems.
- Switch representation: algebra ↔ geometry ↔ trig ↔ complex plane.
- Use symmetry before computation.
- Try modular arithmetic before expanding huge integers.
- Complement before complicated direct counting.

Core algebra identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

Symmetric-expression engine

If $S = a + b$ and $P = ab$:

$$a^2 + b^2 = S^2 - 2P$$

$$a^3 + b^3 = S^3 - 3PS$$

$$(a - b)^2 = S^2 - 4P$$

ADVANCED ALGEBRA

Polynomials, logs, functions & sequences

The late-paper algebra toolkit is mostly about avoiding unnecessary solving.

Quadratics & Vieta CORE

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

$$r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}$$

Optimization trigger: real solutions often imply $\Delta \geq 0$.

Cubic Vieta HIGH

For $ax^3 + bx^2 + cx + d$ with roots r, s, t :

$$r + s + t = -\frac{b}{a}$$

$$rs + rt + st = \frac{c}{a}$$

$$rst = -\frac{d}{a}$$

Remainder / factor theorem

$$\text{remainder of } f(x) \div (x - a) = f(a)$$

$$f(a) = 0 \iff (x - a) \text{ is a factor}$$

Never perform long division if only the remainder is requested.

Rational-root filter

If p/q in lowest terms is a root of an integer polynomial with leading coefficient a_n and constant a_0 :

$$p \mid a_0, \quad q \mid a_n$$

Logs & exponentials CORE

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b(x/y) = \log_b x - \log_b y$$

$$\log_b(x^k) = k \log_b x$$

$$\log_b x = \frac{\log_a x}{\log_a b}$$

Domain: log argument > 0 ; base > 0 , base $\neq 1$.

Standard sums HIGH

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Sequences / telescoping

$$a_n = a_1 + (n-1)d, \quad S_n = \frac{n}{2}(a_1 + a_n)$$

$$S_n^{\text{geom}} = a_1 \frac{1 - r^n}{1 - r}$$

$$S_\infty = \frac{a_1}{1 - r} \quad (|r| < 1)$$

Telescoping trigger: consecutive factors such as $k(k+1)$.

Functions & graphs CORE

$$(f \circ g)(x) = f(g(x)), \quad f^{-1}(f(x)) = x$$

- Domain = allowed inputs; range = outputs.
- $f(x-h) + k$: right h , up k ; $-f(x)$ and $f(-x)$ are reflections.
- Intersections solve $f(x) = g(x)$. For piecewise functions, choose the branch first.

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1, \quad \{x\} = x - \lfloor x \rfloor$$

TRIGONOMETRY

Unit circle, identities & triangle tools

AMC 12 trig is high-value because it links algebraic identities to geometry.

Unit-circle values CORE

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
30°	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	1	0	undef.

Know signs in all four quadrants.

Fundamental identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x, \quad \tan x = \frac{\sin x}{\cos x}$$

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x$$

Sum / difference CORE

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Double angle

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Half angle HIGH

$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$$

$$\cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}$$

Product-to-sum / triple angle BONUS

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

Radians • arcs • sectors CORE

$$180^\circ = \pi \text{ rad}$$

When θ is in **radians**:

$$s = r\theta, \quad A_{\text{sector}} = \frac{1}{2} r^2 \theta$$

Degree formulas use the fraction $\theta/360^\circ$; do not mix the two conventions.

TRIANGLE GEOMETRY

Area, similarity, cevians & radius formulas

These formulas form a network: sides ↔ angles ↔ area ↔ inradius ↔ circumradius.

Triangle formula family CORE

$$K = \frac{1}{2}bh$$

$$K = \frac{1}{2}ab \sin C$$

$$K = rs$$

$$K = \frac{abc}{4R}$$

Here $s = (a + b + c)/2$, r is inradius and R is circumradius.

Heron

$$s = \frac{a + b + c}{2}$$

$$K = \sqrt{s(s-a)(s-b)(s-c)}$$

Best when all three side lengths are known and the altitude is not.

Law of Sines CORE

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

The $2R$ extension is a powerful bridge to circle geometry.

Law of Cosines CORE

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Use for SAS, SSS, or whenever Pythagorean theorem "almost" applies.

Similarity / area ratios

$$\frac{[T_1]}{[T_2]} = \frac{b_1}{b_2} \quad (\text{same altitude})$$

$$\text{side scale } k \Rightarrow \text{area scale } k^2$$

A side divided in ratio $m : n$ gives adjacent triangle areas in ratio $m : n$.

Centroid / angle bisector

$$AG : GM = 2 : 1$$

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Median → equal areas. Angle bisector → side ratio → area ratio.

Stewart / median length HIGH

If cevian d splits side $a = m + n$:

$$b^2m + c^2n = a(d^2 + mn)$$

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Ceva HIGH

$$\frac{BD}{DC} \frac{CE}{EA} \frac{AF}{FB} = 1$$

Use when three cevians are concurrent.

CIRCLE, COORDINATE & 3D GEOMETRY

Power products, cyclic structure and analytic shortcuts

Choose the representation that exposes the fewest unknowns.

Circle angle facts

- Central angle = intercepted arc.
- Inscribed angle = half intercepted arc.
- Same chord → equal inscribed angles.
- Cyclic quadrilateral: opposite angles supplementary.
- Radius $\perp$ tangent at tangency.

Power of a Point CORE

$$PA \cdot PB = PC \cdot PD$$

$$PT^2 = PA \cdot PB$$

Secant: external $\times$ whole = external $\times$ whole. **Tangent-secant:** tangent² = external $\times$ whole.

Ptolemy HIGH

For cyclic quadrilateral $ABCD$:

$$AC \cdot BD = AB \cdot CD + BC \cdot AD$$

Excellent when many side/diagonal lengths are known.

Coordinate formulas

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(x - h)^2 + (y - k)^2 = r^2$$

Point-to-line / shoelace HIGH

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

$$K = \frac{1}{2} \left| \sum x_i y_{i+1} - \sum y_i x_{i+1} \right|$$

Regular polygons

$$\text{interior sum} = (n - 2)180^\circ$$

$$\text{diagonals} = \frac{n(n - 3)}{2}$$

$$\text{exterior angle} = \frac{360^\circ}{n}$$

3D essentials

$$V_{\text{prism}} = Bh, \quad V_{\text{pyramid}} = \frac{1}{3}Bh$$

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad S_{\text{sphere}} = 4\pi r^2$$

$$d_{3D} = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Geometry mode switch

- Cevians/ratios → area + similarity.
- Sides and angles → trig.
- Chords/secants → power of a point.
- Cyclic quadrilateral → Ptolemy.
- Right angles/symmetry → coordinates.

COMPLEX NUMBERS

Algebra and geometry in one language

Complex numbers often turn rotations and regular polygons into one-line algebra.

Basic form CORE

$$z = a + bi, \quad i^2 = -1, \quad \bar{z} = a - bi$$

$$|z| = \sqrt{a^2 + b^2}, \quad z\bar{z} = |z|^2$$

$$|z_1 - z_2| = \text{distance between } z_1 \text{ and } z_2$$

Division / reciprocal

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} \quad (z \neq 0)$$

Multiply numerator and denominator by the conjugate to remove i from the denominator.

Polar form HIGH

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

$$|zw| = |z||w|, \quad \arg(zw) = \arg z + \arg w$$

- Multiplying by i rotates 90° .
- Multiplying by $e^{i\theta}$ rotates by θ .

De Moivre

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

Use for high powers and roots.

Roots of unity HIGH

$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1$$

$$1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0 \quad (\omega \neq 1, \omega^n = 1)$$

Geometrically these are vertices of a regular n -gon on the unit circle.

Conjugate-root fact

If a polynomial has real coefficients:

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

This can determine missing roots without solving the polynomial.

NUMBER THEORY

Prime exponents, congruences & large powers

The late AMC 12 number-theory move is often to choose the right modulus before doing any arithmetic.

Divisors / gcd / lcm CORE

$$n = \prod p_i^{e_i} \Rightarrow \tau(n) = \prod (e_i + 1)$$

$$\sigma(n) = \prod_i \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab$$

- Odd number of divisors iff perfect square.
- Euclid: $\gcd(a, b) = \gcd(b, a \bmod b)$.

Modular arithmetic CORE

$$a \equiv b \pmod{m} \iff m \mid (a - b)$$

- Add, subtract, multiply congruences.
- **Cancellation warning:** cancelling a factor requires appropriate invertibility; it is safe when the cancelled factor is coprime to the modulus.

Quadratic residues

- Squares mod 4: 0, 1.
- Squares mod 5: 0, 1, 4.
- Squares mod 8: 0, 1, 4.
- Every odd square is $1 \pmod{8}$.

Fermat / Euler HIGH

$$a^{p-1} \equiv 1 \pmod{p} \quad (p \text{ prime}, p \nmid a)$$

$$\phi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$$

$$a^{\phi(n)} \equiv 1 \pmod{n} \quad (\gcd(a, n) = 1)$$

Linear congruence / CRT

$$ax \equiv b \pmod{n} \text{ solvable} \iff \gcd(a, n) \mid b$$

If moduli are coprime, simultaneous congruences combine to a unique class modulo their product.

Factorials / valuations HIGH

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \dots$$

Trailing zeros in base 10 = $\min(v_2, v_5)$, usually controlled by v_5 .

Diophantine / SFFT

$$ax + by = c \text{ has integer solutions} \iff \gcd(a, b) \mid c$$

$$xy + ax + by = c \Rightarrow (x + b)(y + a) = c + ab$$

Power-cycle strategy HIGH

- Reduce exponents using an observed cycle, multiplicative order, Fermat or Euler.
- Last two digits can often be split using mod 4 and mod 25.
- Never expand a huge power unless the modulus forces you to.

$$\gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1 \quad (a > 1)$$

BONUS A useful exponential-GCD recognition identity.

COMBINATORICS & PROBABILITY

Selections, arrangements, distributions and expectation

Decide first what counts as a distinct outcome; only then choose a formula.

Permutations / combinations CORE

$$P(n, r) = \frac{n!}{(n-r)!}, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n}{r} = \binom{n}{n-r}, \quad \frac{n!}{a!b!c!\dots}$$

Order matters → permutation. Repeated objects → divide by repeat-factorials. Circular arrangements of n distinct objects: $(n-1)!$ when rotations are identical.

Binomial identities HIGH

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1}$$

Last line = hockey-stick identity.

Stars & bars HIGH

$$x_1 + \dots + x_k = n, \quad x_i \geq 0 : \binom{n+k-1}{k-1}$$

$$x_1 + \dots + x_k = n, \quad x_i \geq 1 : \binom{n-1}{k-1}$$

For lower bounds $x_i \geq a_i$, shift $y_i = x_i - a_i$.

Probability • complement • overlap

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$P(E^c) = 1 - P(E)$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

"At least one" → complement. Conditional multiplication: $P(A \cap B) = P(B)P(A | B)$.

Expected value HIGH

$$E[X] = \sum x_i P(X = x_i)$$

$$E[X + Y] = E[X] + E[Y]$$

Nuance: linearity of expectation does not require independence.

Statistics

$$\mu = E[X], \quad \text{Var}(X) = E[X^2] - E[X]^2$$

If $Y = aX + b$, then $E[Y] = aE[X] + b$ and $\text{Var}(Y) = a^2 \text{Var}(X)$.

Pigeonhole / grid paths

$$\text{some box has at least } \left\lceil \frac{N}{k} \right\rceil$$

$$\text{shortest paths} = \binom{r+u}{r}$$

OPTIMIZATION & POWER TOOLS

Inequalities, invariants and representation switches

AMC 12 forbids calculus, so extrema are usually encoded algebraically.

AM–GM CORE

$$\frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n}$$

Equality iff all terms are equal.

$$x + \frac{1}{x} \geq 2 \quad (x > 0)$$

Fixed sum → product tends to maximize at equality; fixed product → sum tends to minimize at equality.

Cauchy / Titu BONUS

$$(\sum a_i^2)(\sum b_i^2) \geq (\sum a_i b_i)^2$$

$$\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i} \quad (b_i > 0)$$

Discriminant optimization

Convert a constraint into a quadratic in one variable. Real feasibility requires

$$\Delta \geq 0$$

Often produces the exact allowed range of a parameter without calculus.

Complete square / vertex

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}$$

The square term gives the minimum/maximum immediately depending on the sign of a .

Invariant / monovariant

- **Invariant:** unchanged under every legal move.
- **Monovariant:** always moves one direction.
- Typical invariants: parity, sum mod m , color parity, product/sign.

Representation switches HIGH

- Symmetric roots → Vieta.
- Huge powers → modular cycles.
- Cyclic quadrilateral → Ptolemy.
- Chords/tangents → Power of a Point.
- Side-angle geometry → trig.
- Rotation / regular polygon → complex numbers.
- Messy restrictions → complement / inclusion–exclusion.

Normalization / homogeneity

If every term has the same total degree, the expression may depend only on ratios. Set a convenient scale such as $x + y = 1$, $y = 1$, or a side length equal to 1 — **only when the problem is scale-invariant**.

Root-of-unity filter BONUS

When exponents are grouped by residue classes modulo n , roots of unity can filter terms. At AMC 12 level, recognition is more valuable than memorizing a general formula.

FINAL REVIEW

Formula sprint + late-paper traps

Fast recall page for the final minutes before a timed set.

AMC 12-only + algebra sprint

- Vieta: $r + s = -b/a$, $rs = c/a$; remainder by $x - a$: $f(a)$.
- $\log_a(xy) = \log_a x + \log_a y$, $\log_a b = 1/\log_b a$.
- $\sin^2 x + \cos^2 x = 1$, $\sin 2x = 2 \sin x \cos x$.
- Radians: $s = r\theta$, $A_{\text{sector}} = \frac{1}{2}r^2\theta$.
- Law of Sines: $a/\sin A = 2R$; Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos C$.
- Complex: $z\bar{z} = |z|^2$, $|z_1 - z_2| = \text{distance}$.
- De Moivre: $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$.

Geometry sprint

- $K = \frac{1}{2}ab \sin C = rs = abc/(4R)$
- Heron: $K = \sqrt{s(s-a)(s-b)(s-c)}$
- Power: $PT^2 = PA \cdot PB$
- Ptolemy: diagonal product = sum of opposite-side products.
- Centroid ratio: 2:1.
- Angle bisector: adjacent side split = adjacent side ratio.

Number / counting sprint

- $\tau(n) = \prod (e_i + 1)$
- $\gcd \cdot \text{lcm} = ab$
- Fermat: $a^{p-1} \equiv 1 \pmod{p}$
- $\binom{n}{r} = n!/[r!(n-r)!]$
- Stars & bars: $\binom{n+k-1}{k-1}$
- Complement: $1 - P(E^c)$
- Linearity: $E[X + Y] = E[X] + E[Y]$

Common traps HIGH

- $\sqrt{x^2} = |x|$.
- Log arguments must be positive.
- Infinite geometric sum requires $|r| < 1$.
- Squaring can introduce extraneous roots.
- Modular cancellation is not automatic.
- Side scale $k \rightarrow \text{area } k^2, \text{ volume } k^3$.
- Expected-value linearity does not require independence; probability multiplication generally does.

Late-question trigger map

- Repeated polynomial-root expressions $\rightarrow$ Vieta.
- Huge exponent $\rightarrow$ order / Fermat / Euler.
- Triangle + circumcircle $\rightarrow$ extended sine law.
- Cyclic quadrilateral $\rightarrow$ Ptolemy.
- Secant/tangent products $\rightarrow$ Power of a Point.
- Rotation $\rightarrow$ complex multiplication.
- Fixed sum/product extremum $\rightarrow$ AM-GM.
- Feasibility of a quadratic $\rightarrow$ discriminant.

Final checklist

- ☐ Did I use every domain restriction?
- ☐ Did I choose the easiest representation?
- ☐ Can symmetry or a modulus eliminate choices?
- ☐ Did I accidentally assume the diagram is to scale?
- ☐ Did I check extraneous roots?
- ☐ Is there a shorter answer-choice route?